

INTERCEPT HTTP/HTTPS TRAFFIC – Mobile Se Network Hacking

Mobile application penetration testing mein sab se important step hai HTTP aur HTTPS traffic ko intercept karna. Is process mein aap ek proxy tool (jaise Burp Suite) use karte hain jo mobile device aur internet ke beech mein aa kar saari requests aur responses ko capture karta hai . Yeh aapko mobile app ke real-time behavior aur data exchanges ko deep insight ke saath analyze karne ki facility deta hai.

Step 1: Burp Proxy Listener Configure Karein

Sab se pehle Burp Suite ko configure karein taake woh mobile device se aane wali traffic ko accept kar sake.

Burp Suite open karein, Proxy tab par jayein, Proxy Settings mein jaake Proxy Listeners section mein Add button click karein. Binding tab mein Bind to port mein koi bhi available port use karein (recommended: 8080) aur Bind to address mein "**All interfaces**" select karein . Yeh zaroori hai kyunke mobile device Burp ke localhost (127.0.0.1) ko access nahi kar sakta. OK click karein aur confirm karein.

 **Important:** Agar aap "Loopback only" select karein gi toh sirf localhost traffic capture hogi. Mobile device se traffic capture karne ke liye Burp ko saare network interfaces par sunna hoga .

Step 2: Mobile Proxy Settings Configure Karein

Ab mobile device ko configure karein taake woh apna traffic Burp Suite ke through bheje .

Physical Android Device: Settings → Network & Internet → Wi-Fi mein jaayein. Apne connected Wi-Fi network ko long press karein ya uske edit icon par click karein. Advanced options expand karein. Proxy ko "**Manual**" par set karein. Proxy hostname mein apne **PC ki IP address** daalein (192.168.x.x) aur Proxy port mein woh port daalein jo aapne Burp mein set kiya tha (8080) . Save karein aur Wi-Fi se reconnect karein.

iOS Device: Settings → Wi-Fi → [your network] mein jaayein. Network ke ⓘ icon par tap karein, scroll down karein aur HTTP Proxy ko Manual par set karein. Server mein laptop ki IP aur Port mein Burp listener port daalein .

 **IP Address Kaise Pata Karein:** Windows par Command Prompt mein ipconfig type karein aur IPv4 Address dekhein. macOS/Linux par Terminal mein ifconfig ya ip addr type karein .

Step 3: Burp CA Certificate Install Karein (HTTPS Ke Liye)

HTTPS traffic intercept karne ke liye mobile device ko Burp Suite ke CA certificate par trust karna hoga .

◆ **Android (No Root – Limited)**

Mobile browser mein <http://burpsuite> visit karein (HTTPS nahi) aur certificate download karein . Settings → Security → Encryption & credentials → Install from storage mein jaake certificate install karein. **Yeh method sirf browser traffic aur un apps ke liye kaam karegi jinmein SSL Pinning enable nahi hai .**

◆ **Android (Root Required – Full Interception)**

Android 7 (Nougat) se user-installed certificates ko apps trust nahi karti. Isliye HTTPS traffic intercept karne ke liye certificate ko **system store** mein install karna hota hai, jiske liye root access zaroori hai .

Method 1: Manual (Rooted Device)

- Burp certificate ko .der format mein export karein
- PEM format mein convert karein:

```
openssl x509 -inform der -in burp.der -out burp.pem
```
- Certificate hash generate karein:

```
openssl x509 -inform PEM -subject_hash_old -in burp.pem |
```

head -1

(Output: 9a5ba575)

- Certificate ko rename karein: `mv burp.pem 9a5ba575.0`
- ADB se device par push karein aur system certificate directory mein move karein: `adb root`
`adb remount`
`adb push 9a5ba575.0 /sdcard/`
`adb shell`
`su`
`mv /sdcard/9a5ba575.0`
`/system/etc/security/cacerts/`
`chmod 644`
`/system/etc/security/cacerts/9a5ba575.0`

Method 2: Magisk Module (Recommended – Safe)

Agar Magisk se root kiya hai toh

"ConscryptTrustUserCerts" module use karein .

Certificate ko user store mein install karein (as above).

Magisk app mein Modules → Install from storage se module ZIP install karein aur reboot karein. Module automatically user certificates ko system store mein copy kar dega har boot par .

 **Android 14+ Special Case:** Android 14 mein certificate store ko APEX container mein move kar diya gaya hai jise directly modify nahi kiya ja sakta . Magisk module (ConscryptTrustUserCerts) is kaam ke liye zaroori hai .

◆ iOS (iPhone/iPad)

iPhone browser mein <http://burpsuite> visit karein aur certificate download karein . Settings → General → VPN & Device Management → Downloaded Profile → Portswigger CA install karein . Phir Settings → General → About → Certificate Trust Settings mein Portswigger CA ke liye full trust enable karein .

☑ Step 4: Configuration Test Karein

Burp Suite mein Proxy → Intercept tab par jaayein aur Intercept ON karein . Mobile device ka browser open karein aur kisi bhi HTTPS site (jaise <https://example.com>) par jaayein . Agar certificate installation sahi hai toh page load ho jayega bina security warning ke aur Burp Suite mein Proxy → HTTP history tab mein traffic dikhna chahiye .

🚀 Advanced: SSL Pinning Bypass

Bohot saari apps **SSL Pinning** use karti hain jo third-party certificates ko reject kar deti hain . Un apps ka traffic intercept karne ke liye Frida ka use karein :

```
pip install frida-tools
adb push frida-server /data/local/tmp/
adb shell "chmod 755 /data/local/tmp/frida-server
&& /data/local/tmp/frida-server &"
```

```
frida-ps -U  
frida -U -n <app_package_name> -l frida-android-  
unpinning.js
```

Pre-built Frida scripts [GitHub](#) se download kar sakte hain .

Common Issues aur Solutions

Browser SSL Warning Dikha Raha Hai: Certificate sahi se install nahi hua hai. Dobara install karein .

Mobile Par Internet Nahi Chal Raha: Proxy settings galat hain ya Burp Suite ka listener active nahi hai. Ensure karein ke Burp Suite mein proxy listener "All interfaces" par set hai .

Android 14+ Par Certificate Install Nahi Ho Raha:

Android 14+ mein system certificate store ko APEX container mein move kar diya gaya hai, jise modify nahi kiya ja sakta . Solution: Magisk module (ConscryptTrustUserCerts) use karein .

Apps Kaam Nahi Kar Rahi (SSL Pinning): SSL Pinning bypass ke liye Frida script use karein .

Important Security Note

Yeh setup **sirf educational aur authorized penetration testing** ke liye hai . Rooting device se warranty void ho sakti hai aur device security model compromised ho jata hai . Kisi doosre ke network ya device par bina permission ke traffic intercept karna applicable cyber laws ke under punishable hai .

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